

Group17Turtlebot3proj

Changeable Parameter Reference

TurtleBot3 line-follower / barcode-navigation / obstacle-toppling controller. Every tuning constant lives in `__init__`. Parameters with a green environment-variable name can be overridden at launch without editing the file (e.g. `FORWARD_SPEED=0.12 python3 V26_2.py`); parameters marked — must be changed directly in the source.

Runtime environment variables (docstring-level)

Set before launch, e.g. `FORWARD_SPEED=0.12 python3 V26_2.py`. Any CAPITALS parameter built with `os.environ.get()` below can be overridden the same way.

Parameter	Default	Env. var.	What it controls
CONSOLE_STYLE	friendly	<code>CONSOLE_STYLE</code>	Console verbosity: 'friendly' (short, plain-English, one line/sec) or 'debug' (every number).
NO_COLOUR	0	<code>NO_COLOUR</code>	Set to 1 to disable ANSI colour in console output (e.g. when piping to a log file).
ENABLE_ACTUATOR	1	<code>ENABLE_ACTUATOR</code>	Set to 0 for a dry run: full routine runs but the actuator is never powered.
FAIL_STOP_HTML	1	<code>FAIL_STOP_HTML</code>	Set to 0 to stop writing the HTML fail-stop report.

GPIO pin assignments (BCM numbering)

Hard-coded pin numbers -- no environment override. Edit the source to change.

Parameter	Default	Env. var.	What it controls
LEFT_LED	23	—	GPIO pin driving the left indicator LED.
RIGHT_LED	24	—	GPIO pin driving the right indicator LED.
STOP_LED	25	—	GPIO pin driving the stop/red LED.
SERVO_PIN	18	—	GPIO pin driving the linear actuator (PWM, 50 Hz).

LEDs and buzzer

Parameter	Default	Env. var.	What it controls
BLINK_INTERVAL	0.30 s	—	Blink period for the ordinary lost-line alarm.
FAIL_STOP_BLINK_INTERVAL	0.07 s	<code>FAIL_STOP_BLINK_INTERVAL</code>	Very fast blink period used only for a latched fail-stop.

Parameter	Default	Env. var.	What it controls
FAIL_STOP_REMINDER_SEC	15.0 s	<code>FAIL_STOP_REMINDER_SEC</code>	How often the fail-stop recovery reminder is reprinted.
WRITE_FAIL_STOP_HTML	True	<code>FAIL_STOP_HTML</code>	Whether the fail-stop card is also written as an HTML file.
FAIL_STOP_HTML_PATH	~/turtlebot3_fail_stop.html	<code>FAIL_STOP_HTML_PATH</code>	Path the HTML fail-stop report is written to.
TURN_LED_THRESHOLD	0.12	—	Line error magnitude above which the turn-direction LED lights.
BUZZER_INTERVAL	0.90 s	—	Minimum spacing between queued buzzer beeps.
BUZZER_SOUND_VALUE	1	—	TurtleBot3 Sound-service value used as the base buzzer tone.
FAIL_STOP_BUZZER_REPEAT_SEC	2.0 s	<code>FAIL_STOP_BUZZER_REPEAT_SEC</code>	How often the fail-stop siren pattern repeats.
BEEP_BARCODE	{0:[1],1:[3],2:[4,4],3:[5,5]}	—	Beep pattern played for each decoded barcode value.
BEEP_RESCAN	[2, 2]	—	Beep pattern played when the obstacle routine re-scans.
BEEP_NAN_CLOSE_BOARD	[2,5,2,5]	—	Beep pattern for the close-board (too-near) NaN condition.
BEEP_FRONT_NAN_REVERSE	[5,2,5,2]	—	Beep pattern played while reversing off a too-close board.
BEEP_SIDE_CONFIRMED	[4]	—	Beep pattern once the obstacle side (left/right) is confirmed.
BEEP_ALIGNED	[5]	—	Beep pattern once fine alignment on the board is achieved.

Parameter	Default	Env. var.	What it controls
BEEP_TACKLE	[5,4,5]	—	Beep pattern at the start of the push/tackle manoeuvre.
BEEP_SUCCESS	[4, 5]	—	Beep pattern for a successfully completed obstacle routine.
BEEP_FAILURE	[3,3,3]	—	Beep pattern for a recoverable failure.
BEEP_FAIL_STOP	[1,2,3,4,5,4,3,2,1,5,1,5]	—	Rising/falling siren reserved exclusively for a latched fail-stop.

Pixy2 image geometry

The Pixy2 line-tracking image is 79x52 'pixels'.

Parameter	Default	Env. var.	What it controls
FRAME_CENTER_X	39.0 px	—	Column where the line should sit; anything else is steering error.
TRACK_Y	45.0 px	—	Image row (near the bottom) at which the steering error is measured.
PIXY_LAMP_UPPER	1	<code>PIXY_LAMP_UPPER</code>	Pixy2 upper illumination LED on/off.
PIXY_LAMP_LOWER	1	<code>PIXY_LAMP_LOWER</code>	Pixy2 lower illumination LED on/off. Try 0 if glossy tape causes glare.

Normal line-following motion

Speeds are in metres/second unless noted.

Parameter	Default	Env. var.	What it controls
FORWARD_SPEED	0.16	<code>FORWARD_SPEED</code>	Normal line-following forward speed.
ARMED_FORWARD_SPEED	0.15	<code>ARMED_FORWARD_SPEED</code>	Forward speed once a turn command is armed, waiting for the junction.
CURVE_FORWARD_SPEED	0.14	<code>CURVE_FORWARD_SPEED</code>	Forward speed while taking a bend.
SLOW_FORWARD_SPEED	0.15	<code>SLOW_FORWARD_SPEED</code>	Reduced forward speed used in cautious phases.

Parameter	Default	Env. var.	What it controls
CENTERING_SPEED	0.15	<code>CENTERING_SPEED</code>	Forward speed while re-centring on the line.
CURVE_FULL_SLOW_ERROR_PX	18.0 px	<code>CURVE_FULL_SLOW_ERROR_PX</code>	Line-error magnitude at which curve speed is fully reduced.
TURN_SPEED	0.95 rad/s	<code>TURN_SPEED</code>	Angular speed commanded during a junction turn.
MAX_TURN_SPEED	0.95 rad/s	<code>MAX_TURN_SPEED</code>	Angular speed cap during a junction turn.
kp (LINE_KP)	0.050	<code>LINE_KP</code>	Proportional gain of the line-following PD steering controller.
kd (LINE_KD)	0.012	<code>LINE_KD</code>	Derivative gain of the line-following PD steering controller.
FORWARD_DISTANCE_AFTER_INTERSECTION_M	0.1 m	—	Distance driven forward into a junction before turning.
MOVE_CENTER_TIMEOUT_SEC	3.0 s	<code>MOVE_CENTER_TIMEOUT_SEC</code>	Timeout for driving into the junction centre.
TURN_TARGET_RAD	82 deg	—	Target turn angle for a junction turn (converted from degrees).
TURN_TOLERANCE_RAD	3 deg	—	Acceptable error on a completed junction turn.
MAX_TURN_TIME_SEC	4.0 s	—	Timeout for completing a junction turn.

Pixy vector filtering and intersection detection

Units are Pixy image pixels unless noted. No environment overrides.

Parameter	Default	Env. var.	What it controls
MIN_VECTOR_LENGTH_PX	10.0	—	Shortest line vector accepted as real (rejects noise specks).
MIN_FOLLOW_Y_SPAN_PX	7.0	—	Minimum vertical span for a vector to be used for line-following.
MIN_BOTTOM_Y_PX	18.0	—	Minimum bottom-Y for a vector to count as close to the robot.

Parameter	Default	Env. var.	What it controls
MAX_ACCEPTED_ERROR_PX	38.0	—	Largest steering error accepted from a single frame.
MAX_LINE_X_JUMP_PX	26.0	—	Largest frame-to-frame jump in line X position accepted.
MIN_CROSSING_ANGLE_DEG	35.0	—	Minimum angle between two vectors to count as a crossing/junction.
CROSSING_SEGMENT_TOL_PX	5.0	—	Positional tolerance when matching crossing segments.
INTERSECTION_CENTER_X_TOL_PX	26.0	—	How far off-centre an intersection may be and still be accepted.
INTERSECTION_TRIGGER_Y_MIN	14.0	—	Lower Y bound of the intersection trigger zone.
INTERSECTION_TRIGGER_Y_MAX	51.5	—	Upper Y bound of the intersection trigger zone.
LINE_LOST_FRAMES	8 frames	—	Consecutive empty frames before the line is declared lost.
LINE_REACQUIRE_FRAMES	5 frames	—	Consecutive good frames needed to declare the line reacquired.

Barcode reading and voting

Controls how barcode reads are confirmed and how close together barcodes may be placed.

Parameter	Default	Env. var.	What it controls
BARCODE_VOTE_SAMPLES	3	BARCODE_VOTE_SAMPLES	Number of Pixy barcode reads collected per voting window.
BARCODE_MIN_VOTES	2	BARCODE_MIN_VOTES	Votes one code needs (of BARCODE_VOTE_SAMPLES) to be accepted.
BARCODE_VOTE_WINDOW_SEC	0.30 s	BARCODE_VOTE_WINDOW_SEC	Time window over which barcode votes are collected.
BARCODE_VOTE_SPEED	0.09	BARCODE_VOTE_SPEED	Forward speed while a barcode vote is being collected.
BARCODE_CLEAR_FRAMES	6 frames	—	Empty frames required before a barcode latch clears.
INTERSECTION_CONFIRM_FRAMES	2 frames	—	Consecutive frames needed to confirm an intersection.
INTERSECTION_COOLDOWN_SEC	0.5 s	—	Minimum time between two accepted intersections.
BARCODE_ACCEPT_COOLDOWN	0.8 s	BARCODE_ACCEPT_COOLDOWN	Cooldown before the SAME barcode code may be accepted again.
BARCODE_ACCEPT_COOLDOWN_DIFFERENT	0.0 s	BARCODE_ACCEPT_COOLDOWN_DIFFERENT	Cooldown before a DIFFERENT barcode code may be accepted.
BARCODE_LATCH_CLEAR_DISTANCE_M	0.06 m	BARCODE_LATCH_CLEAR_DISTANCE_M	Distance travelled past a barcode before its same-code latch clears.

Parameter	Default	Env. var.	What it controls
MAX_QUEUED_COMMANDS	2	<code>MAX_QUEUED_COMMANDS</code>	Maximum turn commands that may be queued awaiting their junction.
BARCODE_IGNORE_AFTER_COMMAND_SEC	0.00 s	<code>BARCODE_IGNORE_AFTER_COMMAND_SEC</code>	Blind period after any command is issued (0 = disabled).
POST_OBSTACLE_BARCODE_IGNORE_SEC	0.40 s	<code>POST_OBSTACLE_BARCODE_IGNORE_SEC</code>	Blind period for Barcode 2 only, right after the obstacle routine.
BARCODE2_REARM_CLEAR_FRAMES	12 frames	<code>BARCODE2_REARM_CLEAR_FRAMES</code>	Consecutive empty frames required before Barcode 2 may re-arm.
BARCODE2_REARM_DISTANCE_M	0.10 m	<code>BARCODE2_REARM_DISTANCE_M</code>	Distance from the saved obstacle start point before Barcode 2 may re-arm.
BARCODE2_ZONE_MIN_VOTES	3	<code>BARCODE2_ZONE_MIN_VOTES</code>	Votes needed for a non-2 code inside the Barcode-2 zone (unanimous).
DEBUG_VECTOR_LOG	False	—	Enable verbose per-vector debug logging.

Obstacle-routine geometry (measure on the physical robot)

These four numbers describe THIS robot's physical dimensions and should be measured with a ruler, not guessed.

Parameter	Default	Env. var.	What it controls
STOP_DISTANCE_M	0.20 m	<code>STOP_DISTANCE_M</code>	Distance from the board at which the robot stops before pushing.
LIDAR_TO_FRONT_EDGE_M	0.14 m	<code>LIDAR_TO_FRONT_EDGE_M</code>	LiDAR centre to the actuator tip, retracted (measured = 0.14 m).
ACTUATOR_REACH_M	0.05 m	<code>ACTUATOR_REACH_M</code>	Extra reach of the pusher tip when the actuator is fully extended.
TOPPLE_OVERTRAVEL_M	0.04 m	<code>TOPPLE_OVERTRAVEL_M</code>	Extra travel past first contact to push the board past its tipping point.
REAR_OVERHANG_M	0.14 m	<code>REAR_OVERHANG_M</code>	LiDAR centre to the back of the plate carrying the LED board.

Approach, tackle and return speeds

Parameter	Default	Env. var.	What it controls
APPROACH_SPEED	0.10 m/s	—	Base forward speed while

Parameter	Default	Env. var.	What it controls
			approaching the board.
REVERSE_SPEED	-0.20 m/s	—	Reverse speed while returning to the saved start point.
RETURN_TOLERANCE_M	0.02 m	—	Position tolerance when returning to the saved start point.
REVERSE_SLOW_SPEED	-0.10 m/s	—	Reduced reverse speed near the return target.
REVERSE_SLOW_ZONE_M	0.04 m	—	Distance from the target at which reverse speed is reduced.
REVERSE_STOP_MARGIN_M	0.04 m	—	Extra stopping margin during the return reverse.
REVERSE_LINE_CONFIRM_FRAMES	5 frames	—	Frames of line visibility needed to confirm reacquisition while reversing.
STOP_HOLD_DURATION	0.5 s	—	Time the robot holds stationary at the stop distance.
NUDGE_SPEED	0.06 m/s	—	Small forward speed used for close-range nudging.
BOARD_DOWN_THRESHOLD_M	0.35 m	—	LiDAR distance beyond which the board is considered toppled/gone.
ACTUATOR_EXTEND_HOLD_SEC	3.0 s	ACTUATOR_EXTEND_HOLD_SEC	How long the actuator is held extended.
TACKLE_PUSH_SPEED	0.04 m/s	TACKLE_PUSH_SPEED	Forward speed during the push/tackle manoeuvre.
TACKLE_MIN_PUSH_DISTANCE_M	0.04 m	TACKLE_MIN_PUSH_DISTANCE_M	Minimum push distance before the board can be judged toppled.
TACKLE_MAX_PUSH_DISTANCE_M	0.25 m	TACKLE_MAX_PUSH_DISTANCE_M	Maximum allowed push distance before aborting the push.
TACKLE_BOARD_DOWN_CONFIRM_SCANS	3 scans	TACKLE_BOARD_DOWN_CONFIRM_SCANS	Consecutive scans confirming the board is down.

Parameter	Default	Env. var.	What it controls
TACKLE_CONTACT_HOLD_SEC	0.60 s	<code>TACKLE_CONTACT_HOLD_SEC</code>	Time to hold contact against the board once detected.
TACKLE_TIMEOUT_SEC	20.0 s	<code>TACKLE_TIMEOUT_SEC</code>	Overall timeout for the push/tackle state.
TACKLE_STALL_PROGRESS_M	0.004 m	<code>TACKLE_STALL_PROGRESS_M</code>	Odometry progress below which the robot is considered stalled (pushing).
TACKLE_STALL_CONFIRM_SCANS	4 scans	<code>TACKLE_STALL_CONFIRM_SCANS</code>	Consecutive scans confirming a stall/contact condition.
TACKLE_YAW_HOLD_KP	1.2	<code>TACKLE_YAW_HOLD_KP</code>	Proportional gain holding heading steady during the push.
TACKLE_YAW_HOLD_MAX	0.10 rad/s	<code>TACKLE_YAW_HOLD_MAX</code>	Angular speed cap while holding heading during the push.
RETURN_TIMEOUT_SEC	20.0 s	—	Overall timeout for the return-to-start manoeuvre.

Sensor freshness watchdog

Governs when a sensor interruption pauses motion versus triggers a latched fail-stop.

Parameter	Default	Env. var.	What it controls
SENSOR_STALE_TIMEOUT_SEC	0.75 s	<code>SENSOR_STALE_TIMEOUT_SEC</code>	Sensor data age at which it is considered stale and motion pauses.
SENSOR_RECOVERY_GRACE_SEC	3.0 s	<code>SENSOR_RECOVERY_GRACE_SEC</code>	Grace period allowed for sensors to recover before escalating.
SENSOR_RECOVERY_HEALTHY_CYCLES	3 cycles	<code>SENSOR_RECOVERY_HEALTHY_CYCLES</code>	Healthy watchdog cycles required before motion resumes.
SENSOR_MAX_RECOVERIES	2	<code>SENSOR_MAX_RECOVERIES</code>	Maximum recoveries allowed before a permanent fail-stop is latched.
FAIL_STOP_COMMAND_HOLD_SEC	1.0 s	<code>FAIL_STOP_COMMAND_HOLD_SEC</code>	Hold time applied to the last command around a fail-stop.

Parameter	Default	Env. var.	What it controls
WATCHDOG_PERIOD_SEC	0.05 s	<code>WATCHDOG_PERIOD_SEC</code>	Period of the independent sensor-health watchdog timer.
CONTROL_PERIOD_SEC	0.05 s	<code>CONTROL_PERIOD_SEC</code>	Period of the main Pixy control loop timer (20 Hz).

LiDAR range filtering

The LDS-02's specified minimum range is 0.16 m; readings below it are not treated as valid distances.

Parameter	Default	Env. var.	What it controls
SIDE_MIN_RANGE_M	0.16 m	<code>SIDE_MIN_RANGE_M</code>	Minimum valid side-sector LiDAR range.
SIDE_MAX_RANGE_M	0.60 m	<code>SIDE_MAX_RANGE_M</code>	Maximum valid side-sector LiDAR range.
FRONT_MIN_RANGE_M	0.16 m	<code>FRONT_MIN_RANGE_M</code>	Minimum valid front-sector LiDAR range.
FRONT_MAX_RANGE_M	0.60 m	<code>FRONT_MAX_RANGE_M</code>	Maximum valid front-sector LiDAR range.

Side-scan board-side voting

Determines whether the board is to the robot's left or right.

Parameter	Default	Env. var.	What it controls
SIDE_SCAN_COUNT	5 scans	—	Number of LiDAR scans taken for the left/right vote.
SIDE_REQUIRED_VOTES	3	—	Votes (of <code>SIDE_SCAN_COUNT</code>) required for a side to win.
SIDE_MAX_ATTEMPTS	2	—	Maximum rescan attempts if the vote is inconclusive.
SIDE_BOTH_DISTANCE_MARGIN_M	0.02 m	—	If both sides' distances differ by less than this, treat as tied and rescan.
SIDE_MAX_DISTANCE_SPREAD_M	0.30 m	—	Maximum acceptable spread among side-distance samples.
INITIAL_SCAN_SETTLE_DURATION_SEC	0.40 s	—	Settle time before the first obstacle-side scan.
RESCAN_SETTLE_DURATION_SEC	0.40 s	—	Settle time before a stationary rescan.

Close-board NaN classifier

Detects a board so close the LiDAR reports NaN instead of a distance.

Parameter	Default	Env. var.	What it controls
NAN_SIDE_MIN_RATIO	0.55	<code>NAN_SIDE_MIN_RATIO</code>	Minimum fraction of NaN returns on a side to count as 'board too close'.

Parameter	Default	Env. var.	What it controls
NAN_SIDE_ADVANTAGE	0.30	<code>NAN_SIDE_ADVANTAGE</code>	Required NaN-ratio advantage of one side over the other.
NAN_VOTE_WINDOW	5	<code>NAN_VOTE_WINDOW</code>	Number of scans kept in the NaN-side voting window.
NAN_REQUIRED_VOTES	3	<code>NAN_REQUIRED_VOTES</code>	Votes required within the window to confirm a NaN-side result.

Front NaN handling (board too close after the turn)

Parameter	Default	Env. var.	What it controls
FRONT_NAN_HALF_ANGLE_DEG	25.0 deg	<code>FRONT_NAN_HALF_ANGLE_DEG</code>	Half-angle of the front sector examined for NaN returns.
FRONT_NAN_MIN_RATIO	0.50	<code>FRONT_NAN_MIN_RATIO</code>	Minimum NaN fraction in the front sector to trigger the reverse.
FRONT_NAN_CONFIRM_SCANS	3 scans	<code>FRONT_NAN_CONFIRM_SCANS</code>	Consecutive NaN-heavy scans required before backing away.
FRONT_NAN_EDGE_FALLBACK	True	<code>FRONT_NAN_EDGE_FALLBACK</code>	Treat an off-axis cluster as an edge artefact when the pusher axis itself is clear.
FRONT_NAN_REVERSE_DISTANCE_M	0.10 m	<code>FRONT_NAN_REVERSE_DISTANCE_M</code>	Distance reversed to clear a too-close board.
FRONT_NAN_REVERSE_SPEED	-0.05 m/s	<code>FRONT_NAN_REVERSE_SPEED</code>	Speed while reversing off a too-close board.
FRONT_NAN_RETURN_SPEED	0.05 m/s	<code>FRONT_NAN_RETURN_SPEED</code>	Speed while driving back forward after that reverse.
FRONT_NAN_HEADING_KP	1.5	<code>FRONT_NAN_HEADING_KP</code>	Heading-hold gain during the front-NaN reverse/return.
FRONT_NAN_MAX_ANGULAR_SPEED	0.15 rad/s	<code>FRONT_NAN_MAX_ANGULAR_SPEED</code>	Angular speed cap during the front-NaN reverse/return.
FRONT_NAN_REVERSE_TIMEOUT_SEC	5.0 s	<code>FRONT_NAN_REVERSE_TIMEOUT_SEC</code>	Timeout for the front-NaN reverse manoeuvre.

Rear geometry and rear-NaN handling

Parameter	Default	Env. var.	What it controls
REAR_NAN_REVERSE_MARGIN_M	0.03 m	REAR_NAN_REVERSE_MARGIN_M	Safety margin added to rear clearance before reversing.
REAR_NAN_INCREASE_BLOCK_RATIO	0.20	REAR_NAN_INCREASE_BLOCK_RATIO	NaN-ratio increase over baseline that blocks a reverse.
REAR_BASELINE_MATCH_TOLERANCE_M	0.03 m	REAR_BASELINE_MATCH_TOLERANCE_M	Tolerance for matching a rear reading to the robot's own self-view baseline.

Obstacle 90-degree turn

Stage 1 of the obstacle routine: an odometry-based turn to face the board.

Parameter	Default	Env. var.	What it controls
OBSTACLE_COARSE_TURN_DEG	90.0 deg	OBSTACLE_COARSE_TURN_DEG	Target turn angle to face the obstacle side.
OBSTACLE_TURN_TOLERANCE_RAD	3 deg	OBSTACLE_TURN_TOLERANCE_DEG	Acceptable error on the obstacle-facing turn.
OBSTACLE_TURN_SPEED	0.60 rad/s	OBSTACLE_TURN_SPEED	Angular speed for the bulk of the turn.
OBSTACLE_TURN_SLOW_SPEED	0.28 rad/s	OBSTACLE_TURN_SLOW_SPEED	Reduced angular speed as the turn nears its target.
OBSTACLE_TURN_CREEP_SPEED	0.10 rad/s	OBSTACLE_TURN_CREEP_SPEED	Slowest creep speed right at the end of the turn.
OBSTACLE_TURN_SLOW_ZONE_RAD	22 deg	—	Angle-to-go at which the turn switches to slow speed.
OBSTACLE_TURN_CREEP_ZONE_RAD	8 deg	—	Angle-to-go at which the turn switches to creep speed.
OBSTACLE_TURN_TIMEOUT_SEC	6.0 s	OBSTACLE_TURN_TIMEOUT_SEC	Timeout for completing the obstacle-facing turn.

Front cluster detection and fine alignment

Stage 2: find the board cluster in the LiDAR scan and aim the pusher at it.

Parameter	Default	Env. var.	What it controls
FRONT_CLUSTER_HALF_ANGLE_DEG	45.0 deg	<code>FRONT_CLUSTER_HALF_ANGLE_DEG</code>	Half-angle of the sector searched for the board cluster.
FRONT_CLUSTER_MIN_POINTS	3	<code>FRONT_CLUSTER_MIN_POINTS</code>	Minimum LiDAR points required to count as a valid cluster.
FRONT_CLUSTER_MAX_ANGLE_GAP_DEG	3.0 deg	<code>FRONT_CLUSTER_MAX_ANGLE_GAP_DEG</code>	Maximum angular gap between points still counted as one cluster.
FRONT_CLUSTER_MAX_RANGE_JUMP_M	0.10 m	<code>FRONT_CLUSTER_MAX_RANGE_JUMP_M</code>	Maximum range jump between points still counted as one cluster.
FRONT_CLUSTER_MIN_WIDTH_M	0.015 m	<code>FRONT_CLUSTER_MIN_WIDTH_M</code>	Smallest cluster width accepted as a real board.
FRONT_CLUSTER_MAX_WIDTH_M	0.50 m	<code>FRONT_CLUSTER_MAX_WIDTH_M</code>	Largest cluster width accepted as a real board.
ACTUATOR_AIM_OFFSET_DEG	0.0 deg	<code>ACTUATOR_AIM_OFFSET_DEG</code>	Fixed bearing offset correction for actuator/LiDAR mounting misalignment.
FINE_ALIGN_SETTLE_DURATION_SEC	0.25 s	<code>FINE_ALIGN_SETTLE_DURATION_SEC</code>	Settle time before fine alignment scanning begins.
FINE_ALIGN_TOLERANCE_DEG	5.0 deg	<code>FINE_ALIGN_TOLERANCE_DEG</code>	Legacy centre-seek bearing tolerance (<code>ALIGN_MODE=centre</code>).
FRONT_ALIGN_CONFIRM_SCANS	3 scans	<code>FRONT_ALIGN_CONFIRM_SCANS</code>	Consecutive agreeing scans required to confirm alignment.
FINE_ALIGN_KP	1.8	<code>FINE_ALIGN_KP</code>	Proportional gain of the fine-alignment rotation controller.
FINE_ALIGN_MIN_SPEED	0.045 rad/s	<code>FINE_ALIGN_MIN_SPEED</code>	Minimum rotation speed during fine alignment.
FINE_ALIGN_MAX_SPEED	0.18 rad/s	<code>FINE_ALIGN_MAX_SPEED</code>	Maximum rotation speed during fine alignment.
FINE_ALIGN_TIMEOUT_SEC	8.0 s	<code>FINE_ALIGN_TIMEOUT_SEC</code>	Timeout for the fine-alignment state.

Aim tolerance window (lateral-miss alignment)

The board is aimed at using lateral miss distance on the board face, not a fixed bearing angle. Set `ALIGN_MODE=centre` to restore the old strict centre-seeking behaviour.

Parameter	Default	Env. var.	What it controls
ALIGN_MODE	window	<code>ALIGN_MODE</code>	Alignment strategy: 'window' (lateral-miss, default) or 'centre' (legacy strict).
BOARD_HIT_MARGIN_M	0.030 m	<code>BOARD_HIT_MARGIN_M</code>	Keep-off margin from each board edge inside the hit window.

Parameter	Default	Env. var.	What it controls
BOARD_HIT_WINDOW_MIN_M	0.015 m	<code>BOARD_HIT_WINDOW_MIN_M</code>	Narrowest hit window ever accepted.
BOARD_HIT_WINDOW_MAX_M	0.060 m	<code>BOARD_HIT_WINDOW_MAX_M</code>	Widest hit window ever trusted, even for a wide board.
ALIGN_ENTER_FRACTION	0.60	<code>ALIGN_ENTER_FRACTION</code>	Fraction of the window required to ENTER alignment (hysteresis).
FINE_ALIGN_MAX_ANGLE_DEG	15.0 deg	<code>FINE_ALIGN_MAX_ANGLE_DEG</code>	Hard sanity cap on bearing error regardless of window size.
APPROACH_MAX_ANGLE_DEG	25.0 deg	<code>APPROACH_MAX_ANGLE_DEG</code>	Hard sanity cap on bearing error while approaching.
APPROACH_ROTATE_ONLY_FACTOR	2.0	<code>APPROACH_ROTATE_ONLY_FACTOR</code>	Miss size (in window-widths) at which the robot stops translating and rotates only.
APPROACH_AIM_MAX_SCANS	12 scans	<code>APPROACH_AIM_MAX_SCANS</code>	Scans at the braking distance before accepting any on-board aim as a last resort.
FRONT_SAFETY_STOP_M	0.17 m	<code>FRONT_SAFETY_STOP_M</code>	Absolute closest distance allowed before the robot must stop.

Approach to the board

Parameter	Default	Env. var.	What it controls
APPROACH_NEAR_DISTANCE_M	0.40 m	<code>APPROACH_NEAR_DISTANCE_M</code>	Distance at which the approach switches to near-board speed.
APPROACH_NEAR_SPEED	0.08 m/s	<code>APPROACH_NEAR_SPEED</code>	Forward speed once inside <code>APPROACH_NEAR_DISTANCE_M</code> .
APPROACH_AIM_TOLERANCE_DEG	4.0 deg	<code>APPROACH_AIM_TOLERANCE_DEG</code>	Legacy fixed-angle aim tolerance (<code>ALIGN_MODE=centre</code> only).
APPROACH_ROTATE_ONLY_DEG	9.0 deg	<code>APPROACH_ROTATE_ONLY_DEG</code>	Legacy rotate-only bearing threshold (<code>ALIGN_MODE=centre</code> only).
APPROACH_BALANCE_KP	1.8	<code>APPROACH_BALANCE_KP</code>	Proportional gain correcting bearing while approaching.
APPROACH_MAX_ANGULAR_SPEED	0.20 rad/s	<code>APPROACH_MAX_ANGULAR_SPEED</code>	Angular speed cap while approaching.
APPROACH_TIMEOUT_SEC	10.0 s	<code>APPROACH_TIMEOUT_SEC</code>	Timeout for the approach state.
APPROACH_MAX_DISTANCE_M	0.75 m	<code>APPROACH_MAX_DISTANCE_M</code>	Maximum plausible board distance; anything further is rejected.

Post-obstacle line reacquisition

Parameter	Default	Env. var.	What it controls
OBSTACLE_REACQUIRE_CONFIRM_FRAMES	2 frames	<code>OBSTACLE_REACQUIRE_CONFIRM_FRAMES</code>	Frames of confirmed line needed to leave the reacquire state.
OBSTACLE_REACQUIRE_SPEED	0.13 m/s	<code>OBSTACLE_REACQUIRE_SPEED</code>	Forward speed while reacquiring the line.
OBSTACLE_REACQUIRE_LARGE_ERROR_SPEED	0.10 m/s	<code>OBSTACLE_REACQUIRE_LARGE_ERROR_SPEED</code>	Reduced forward speed when the line error is large during reacquisition.
OBSTACLE_REACQUIRE_MAX_ANGULAR_SPEED	0.70 rad/s	<code>OBSTACLE_REACQUIRE_MAX_ANGULAR_SPEED</code>	Angular speed cap while reacquiring the line.
OBSTACLE_REACQUIRE_SEARCH_SPEED	0.35 rad/s	<code>OBSTACLE_REACQUIRE_SEARCH_SPEED</code>	Angular speed while actively searching (line not yet visible).

Compatibility / legacy fallback

Parameter	Default	Env. var.	What it controls
CENTERING_BALANCE_TOLERANCE	0.060	—	Legacy left/right balance tolerance for the old fallback centring branch.
CENTERING_EXTRA_DURATION_SEC	0.0 s	—	Legacy extra centring duration for the old fallback branch.

Actuator pulse widths

Parameter	Default	Env. var.	What it controls
ACTUATOR_EXTEND_PULSE_MS	2.0 ms	<code>ACTUATOR_EXTEND_PULSE_MS</code>	RC-style PWM pulse width commanding the actuator to extend.
ACTUATOR_RETRACT_PULSE_MS	1.0 ms	<code>ACTUATOR_RETRACT_PULSE_MS</code>	RC-style PWM pulse width commanding the actuator to retract.

Console status logging and LiDAR diagnostics

Parameter	Default	Env. var.	What it controls
STATUS_LOG_INTERVAL	1.00 s	<code>STATUS_LOG_INTERVAL</code>	Minimum interval between printed status lines.
BLIND_VALUE_MODE	nan	<code>BLIND_VALUE_MODE</code>	What counts as a 'blind' LiDAR reading: nan / zero / both.
BLIND_ZERO_EPSILON_M	0.01 m	<code>BLIND_ZERO_EPSILON_M</code>	Threshold below which a reading is treated as zero/blind.
LIDAR_DIAGNOSTIC	False	<code>LIDAR_DIAGNOSTIC</code>	Enable one-off LiDAR diagnostic report at startup.